

Md Mazharul Islam

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 Google Scholar |  ORCID |  Scopus |  GitHub |  LinkedIn

RESEARCH PROFILE

My research focuses on security, privacy, and trustworthy AI for data-driven and distributed systems. My work spans searchable and homomorphic encryption, privacy-preserving data sharing, blockchain-based security, federated learning, and LLM security. I have developed practical privacy-preserving systems for cloud and IoT data, healthcare, and insurance, and more recently investigated differentially private federated learning, Sybil-resilient aggregation, and secure LLM applications in cyber-physical systems.

EDUCATION

- North South University Graduated: Fall 2020
Master of Science in Computer Science and Engineering Dhaka, Bangladesh
◦ CGPA: 3.97 / 4.00.
- North South University Graduated: Summer 2018
Bachelor of Science in Computer Science and Engineering Dhaka, Bangladesh
◦ CGPA: 3.24 / 4.00

PUBLICATIONS

[J] JOURNAL · [C] CONFERENCE · [R] UNDER REVIEW · [T] THESIS

Journal Articles (Published)

- [J.2] **Md Mazharul Islam**, Abrar Mohammed Tanzim Alam, Md. Sadikur Rahman Rony, and Susmay Das, "Data-centric, robust, and explainable multimodal deep learning for clinical decision support: A systematic review," *International Journal of Medical Informatics*, art. 106684, 2026. [doi:10.1016/j.ijmedinf.2026.106684](https://doi.org/10.1016/j.ijmedinf.2026.106684)
- [J.1] **Md Mazharul Islam**, Mubasshir Ahmed, Rajesh Palit, Mohammad Shahriar Rahman, and Salekul Islam, "Fraud detection in privacy preserving health insurance system using blockchain technology," *Engineering Reports*, vol. 7, no. 8, art. e70315, 2025. [doi:10.1002/eng2.70315](https://doi.org/10.1002/eng2.70315)

Peer-Reviewed Conference Papers

- [C.7] **Md Mazharul Islam**, Mubasshir Ahmed, and Niaz Ashraf Khan, "CPS-Guard: Defensive architectures for securing large language models in cyber-physical systems," in *Proc. 28th Int. Conf. on Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, 2025, pp. 2802–2807. [doi:10.1109/ICCIT68739.2025.11491276](https://doi.org/10.1109/ICCIT68739.2025.11491276)
- [C.6] **Md Mazharul Islam**, Mubasshir Ahmed, Shakib Hossain, and Niaz Ashraf Khan, "A blockchain-based medical record storage system for healthcare data management," in *Proc. 4th Int. Conf. on Advances in Communication Technology and Computer Engineering (ICTACE), Lecture Notes in Networks and Systems*, vol. 1312, Springer, 2024, pp. 273–285. [doi:10.1007/978-3-031-94620-2_24](https://doi.org/10.1007/978-3-031-94620-2_24)
- [C.5] **Md Mazharul Islam**, Tahmid Ashraf Khan, Sunjare Zulfiker, A. S. M. Jahid Hasan, and Hafiz Abdur Rahman, "Remotely accessible cyber-physical system testbed for power grid's security and reliability," in *Proc. 8th Int. Conf. on Smart Grid and Smart Cities (ICSGSC)*, Shanghai, China, 2024, pp. 436–441. [doi:10.1109/ICSGSC62639.2024.10813893](https://doi.org/10.1109/ICSGSC62639.2024.10813893)
- [C.4] **Md Mazharul Islam**, Mubasshir Ahmed, and Rajesh Palit, "Fault detection and diagnostics of air handling unit in hospital building using machine learning," in *Proc. 15th Int. Conf. on Computing Communication and Networking Technologies (ICCCNT)*, Kamand, India, 2024, pp. 1–6. [doi:10.1109/ICCCNT61001.2024.10724490](https://doi.org/10.1109/ICCCNT61001.2024.10724490)
- [C.3] **Md Mazharul Islam**, Mubasshir Ahmed, and Rajesh Palit, "Storage and maintenance of sensitive data using homomorphic encryption," in *Proc. 16th Int. Conf. on Security of Information and Networks (SIN)*, Jaipur, India, 2023, pp. 1–6. [doi:10.1109/SIN60469.2023.10474775](https://doi.org/10.1109/SIN60469.2023.10474775)
- [C.2] **Md Mazharul Islam** and Rajesh Palit, "A keyword based searching and sharing scheme on the encrypted cloud data," in *Proc. 14th Int. Conf. on Computing Communication and Networking Technologies (ICCCNT)*, Delhi, India, 2023, pp. 1–6. [doi:10.1109/ICCCNT56998.2023.10308309](https://doi.org/10.1109/ICCCNT56998.2023.10308309)
- [C.1] **Md Mazharul Islam**, Rajib Imran, and Shazzad Hosain, "The evaluation of enterprise resource planning using ISO 25010 based quality model," in *Proc. 2nd Int. Informatics and Software Engineering Conf. (IISEC)*, Ankara, Turkey, 2021, pp. 1–6. [doi:10.1109/IISEC54230.2021.9672349](https://doi.org/10.1109/IISEC54230.2021.9672349)

Manuscripts Under Review

- [R.4] **Md Mazharul Islam**, Abrar Mohammed Tanzim Alam, and Mubasshir Ahmed, "CR-SHE: Collusion-resilient searchable homomorphic encryption for multi-cloud IoT data sharing," *IEEE Internet of Things Journal*. (under review, 2026) [manuscript]
- [R.3] **Md Mazharul Islam**, Mohammad Kaosain Akbar, and Niaz Ashraf Khan, "SyPrFL: Sybil-resilient privacy-preserving federated learning via differentially-private projection clustering," *Journal of Information Security and Applications* (Elsevier). (under review, 2026) [manuscript]
- [R.2] Mubasshir Ahmed, **Md Mazharul Islam**, Rajesh Palit, Mohammad Shahriar Rahman, and Salekul Islam, "SHIELD: Secure hybrid insurance ledgers and ensemble detection for health insurance fraud prevention," *IET Blockchain*. (under review, 2026) [manuscript]
- [R.1] **Md Mazharul Islam**, Md. Shahadul Alam Patwary, Salekul Islam, and Rajesh Palit, "CryptoCloud: A usable framework for storing, searching and sharing encrypted data on cloud storage," *Security and Privacy* (Wiley). (under review, 2026) [manuscript]

Thesis and Preprint

- [T.1] **Md Mazharul Islam**, "A practical framework for storing and searching encrypted data on cloud storage," M.Sc. thesis, Dept. of Electrical and Computer Engineering, North South University, Dhaka, Bangladesh, 2020. Supervisor: Prof. Rajesh Palit. Preprint: [arXiv:2306.03547](https://arxiv.org/abs/2306.03547).

RESEARCH EXPERIENCE

- Institute for Advanced Research (IAR) January 2024 – January 2026
Research Assistant United International University, Dhaka
 - Developed a privacy-preserving health-insurance fraud detection framework combining blockchain-based record integrity with ensemble machine learning for anomaly detection.
 - Developed *SHIELD*, a hybrid insurance-ledger framework integrating permissioned blockchain, privacy-preserving data management, and ensemble fraud detection.
- Cyber-Physical Systems Research Lab (CPS-PMU) February 2022 – April 2024
Research Assistant (PI: Prof. Hafiz Abdur Rahman) North South University, Dhaka
 - Investigated cyber-physical attacks on power grids by building a remotely accessible smart-grid testbed that synchronises GridPACK for power-flow simulation, NS-3 for network modelling, and HELICS for co-simulation, enabling reproducible experiments across coupled cyber and physical layers.
 - Developed Node-RED-based interfaces for remote testbed access and designed repeatable attack scenarios for evaluating the impact of network-level intrusions on power-grid operation and reliability.

HONORS AND AWARDS

- Vice-Chancellor's Gold Medal** December 2025
North South University - highest academic distinction of the MSc CSE programme; conferred at the 25th Convocation
- Huawei ICT Competition 2023–2024, Network Track** May 2024
Global Final, Shenzhen, China - 2nd place as part of a 3-member national team
3rd place, Bangladesh National Round; represented Bangladesh at the Asia-Pacific regional stage, Jakarta, Indonesia.

ACADEMIC SERVICE

- Reviewer**, Journal of Information Security and Applications (Elsevier) 2026
- Reviewer**, Int. Conf. on Data, Computation and Communication (ICDCC), VIT Bhopal, India 2024
- Reviewer**, Int. Conf. on Innovations in Data Science (ICIDS), Manipal Univ. Jaipur, India 2024

TECHNICAL SKILLS

- Cryptography & Privacy**: Searchable encryption; homomorphic encryption (Paillier); secure aggregation; secret sharing; pseudorandom functions; secure multi-party computation; differential privacy; privacy leakage analysis.
- Privacy-Preserving & Secure ML**: Federated learning; differentially private training; Sybil and poisoning attacks; robust aggregation; anomaly detection; explainable AI. *Tools*: PyTorch, scikit-learn, Flower, Opacus.
- LLM Security & Trustworthy AI**: LLM fine-tuning; privacy-preserving training; cybersecurity applications of LLMs; adversarial threat modelling; secure and defensive LLM architectures; trustworthy AI.
- Cyber-Physical & IoT Security**: Cyber-physical co-simulation using NS-3, GridPACK, and HELICS; smart-grid testbed development; attack-scenario design; IoT intrusion detection.
- Research & Implementation**: Python, C++, Bash, SQL; Linux; Git; reproducible experimentation; prototype development; systematic review (PRISMA); $\LaTeX$.
- Cloud & Distributed Systems**: Docker, Kubernetes, Terraform; AWS, OCI; cloud security; IAM; backup and disaster recovery.

TEACHING EXPERIENCE

- Department of Electrical & Computer Engineering July 2023 – June 2024
Teaching Assistant North South University, Dhaka
 - Led supplemental instruction and exam-review sessions for CSE 411 (Database Systems), CSE 422 (Simulation and Modeling), and CSE 482 (Web Technologies), supporting approximately 40 students per semester.
 - Supervised laboratory sessions, graded coursework, and provided individual academic guidance to students.

PROFESSIONAL EXPERIENCE

- Bangladesh Data Center Company Limited (BDCCL)  April 2024 – Present
Assistant Manager (Cloud) Dhaka, Bangladesh
 - Provision and operate Oracle Cloud Infrastructure environments for government and enterprise tenants, managing compute, storage, networking, IAM, and managed database services.
 - Provide technical support to tenants, diagnosing faults across compute, storage, networking, and access management, and maintaining backup and disaster-recovery posture.
 - Prepare architecture notes, standard operating procedures, and handover documentation for tenant onboarding and service delivery.
- Ethics Advanced Technology Limited (EATL)  September 2023 – November 2023
Cyber Security Lab Instructor Dhaka, Bangladesh
 - Delivered hands-on security training to 50+ IT professionals from national banks using EC-Council ICBT Cybersecurity Essentials laboratories, designing lab exercises in network defence, ethical hacking, and incident response.
 - Guided participants through practical security exercises involving vulnerability assessment, attack simulation, and defensive techniques in controlled lab environments.
- One World Infotech Limited  December 2020 – May 2021
Solution Engineer Dhaka, Bangladesh
 - Deployed enterprise authentication and access-control solutions for banking and enterprise clients, including multi-factor authentication (OneSpan), covering technical scoping, proof-of-concept, and production rollout.
 - Conducted threat and vulnerability assessments of client environments and contributed to security operations centre design and incident-response readiness engagements.

CERTIFICATIONS

- **AWS Certified Cloud Practitioner (CLF-C02)** [\[verify\]](#) June 2025
- **AWS Certified AI Practitioner (AIF-C01)** [\[verify\]](#) June 2025
- **Certified Ethical Hacker - Master, ANSI, and Practical** [\[verify\]](#) Dec 2024 – Jan 2025
- **OCI 2024 Certified AI Foundations Associate** [\[verify\]](#) August 2024
- **OCI 2024 Generative AI Certified Professional** [\[verify\]](#) July 2024

REFERENCES

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